

YZM2041

Artificial Intelligence and Ethics Semester Project

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Project Link: (<https://github.com/oykutugana/ai-ethics-aiaaic-analysis>)

Can AI Judge the Ethics of AI?

A Study of Real AI Ethics Incidents, 2021 to 2026

AI and Ethics Semester Project

1. Overview

This project examines real world cases in which artificial intelligence caused ethical concerns. The cases were collected from the AIAAIC database, which records documented AI incidents reported in news and research sources. The central question is whether AI systems are able to recognise and judge the ethics of AI related actions.

The study was carried out in three phases. The first phase describes the structure of AI ethics incidents. The second phase compares how four large language models judge the same cases. The third phase builds a small machine learning model to predict the ethical topic of a case from its title.

2. Data and Method

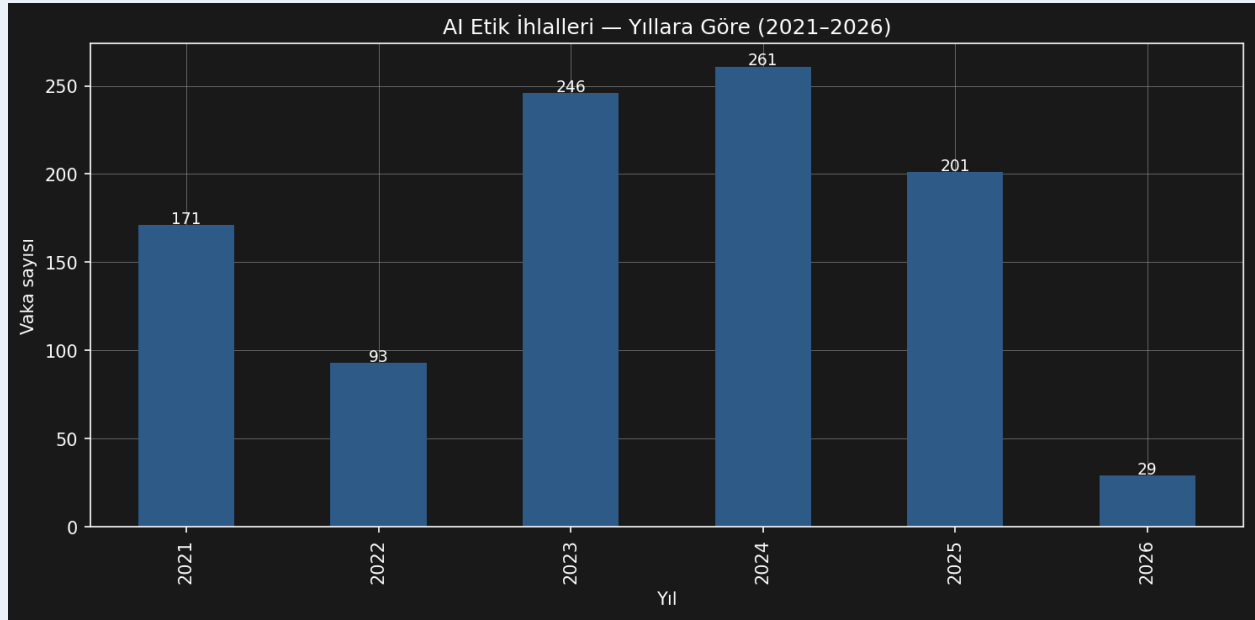
The raw data was cleaned before analysis. Spelling variants were corrected, entries that were not related to AI were removed, and cases that described only technical failures rather than ethical choices were excluded. The period was limited to the years 2021 to 2026. After cleaning, 1001 cases remained.

Each case contains a title, a year, a technology, one or more ethical topics, and a sector. Since a single case can belong to several ethical topics at once, the data was treated as multi labelled throughout the analysis.

3. Phase 1: The Structure of AI Ethics Incidents

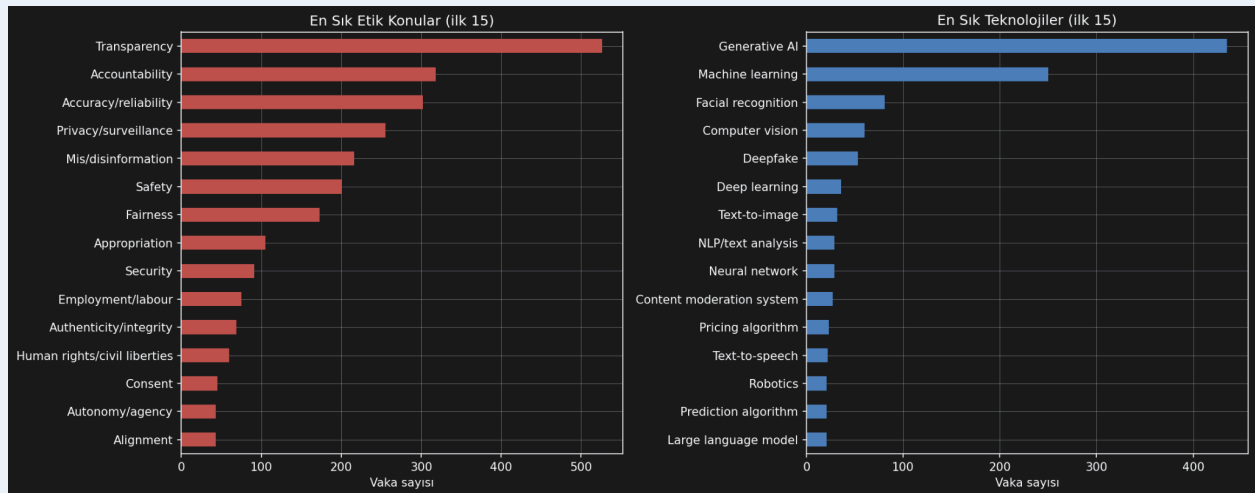
In the first phase, the overall shape of the incidents was explored through trends, frequent topics, technologies, and sectors. The results are summarised below, with each chart placed next to the point it supports.

A clear rise in the number of incidents was observed after 2022, the period in which generative AI became widely available.



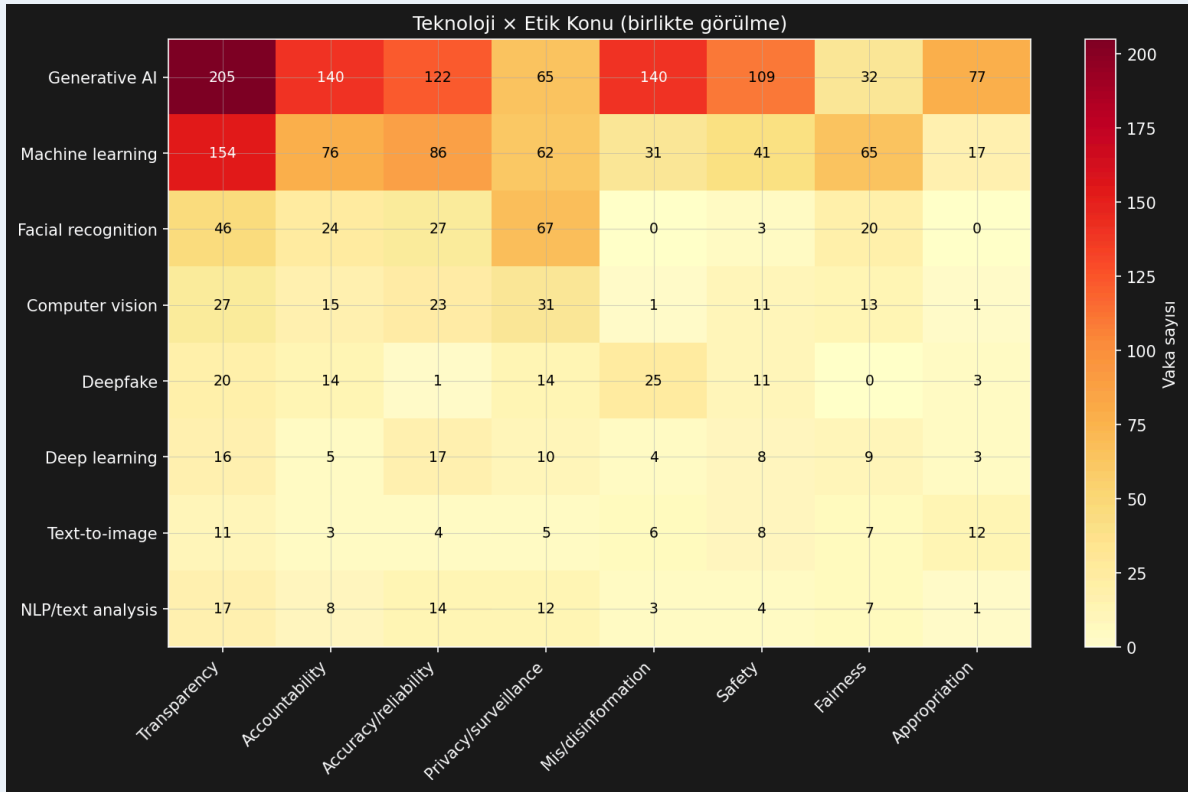
02_yillara_gore_trend.png

Transparency was found to be the most frequent ethical topic, and generative AI the most frequent technology.



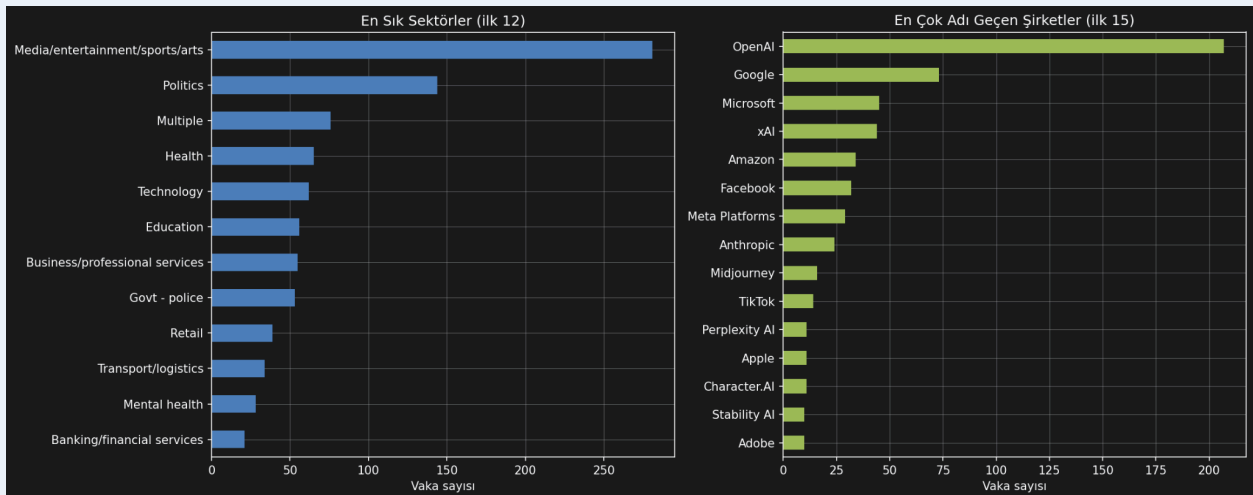
03_konu_teknoloji.png

Each technology tends to be linked to a particular ethical concern. Facial recognition is associated with privacy, while deepfakes are associated with disinformation.



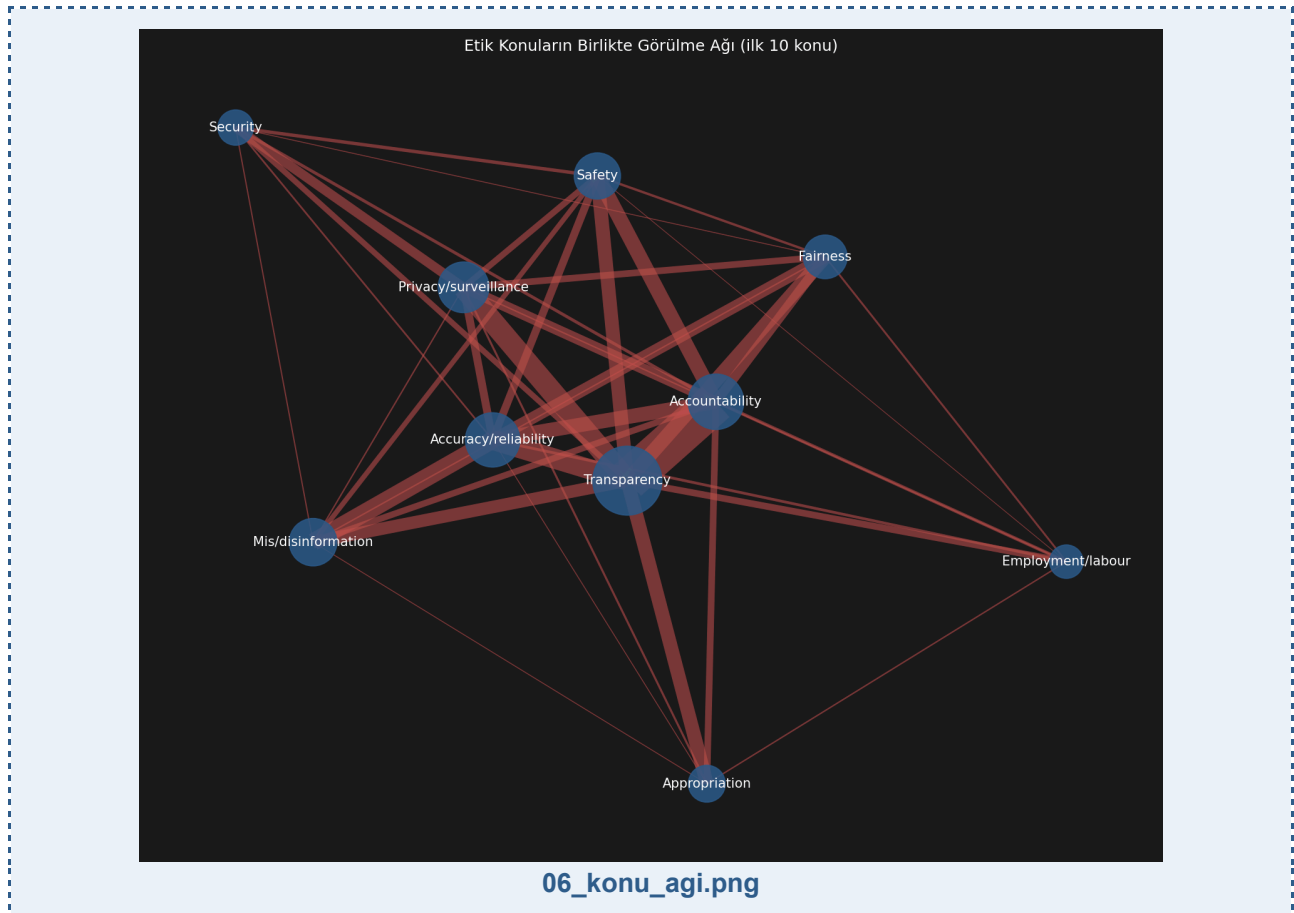
04_teknoloji_konu_isihasiritasi.png

Media and politics were identified as the sectors with the most incidents. OpenAI was the most frequently named developer, which reflects visibility and wide use rather than a judgement of guilt.

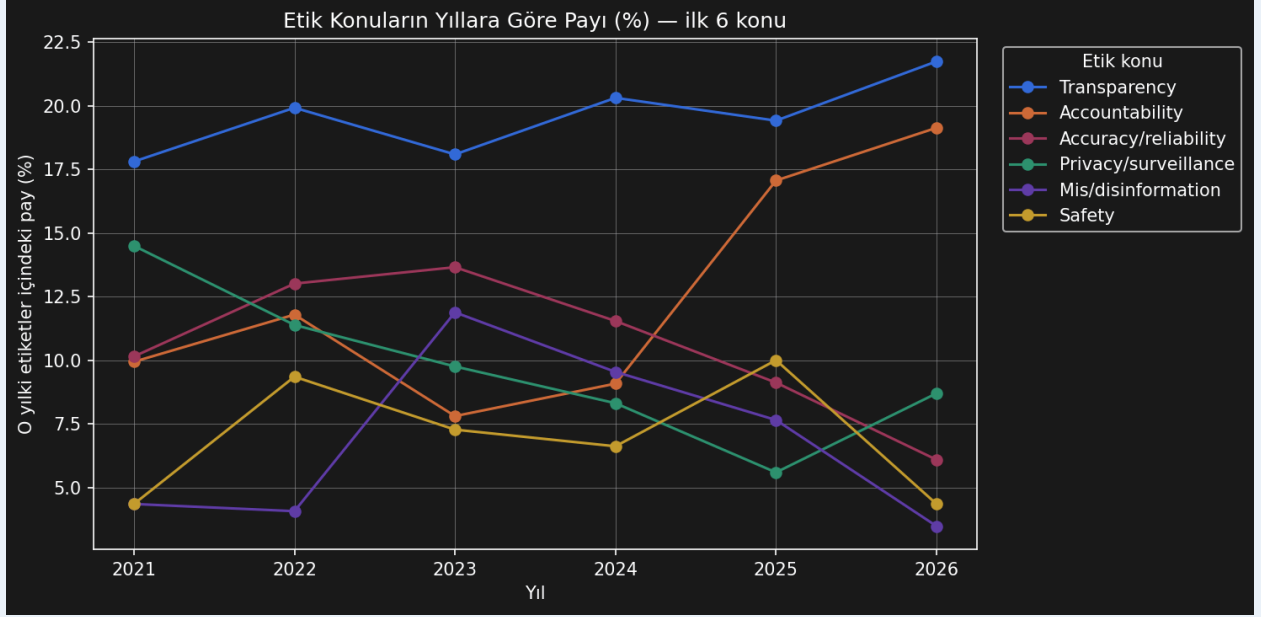


05_sektor_sirket.png

Ethical topics rarely appear alone. Transparency, accountability, and accuracy were found to occur together very often.

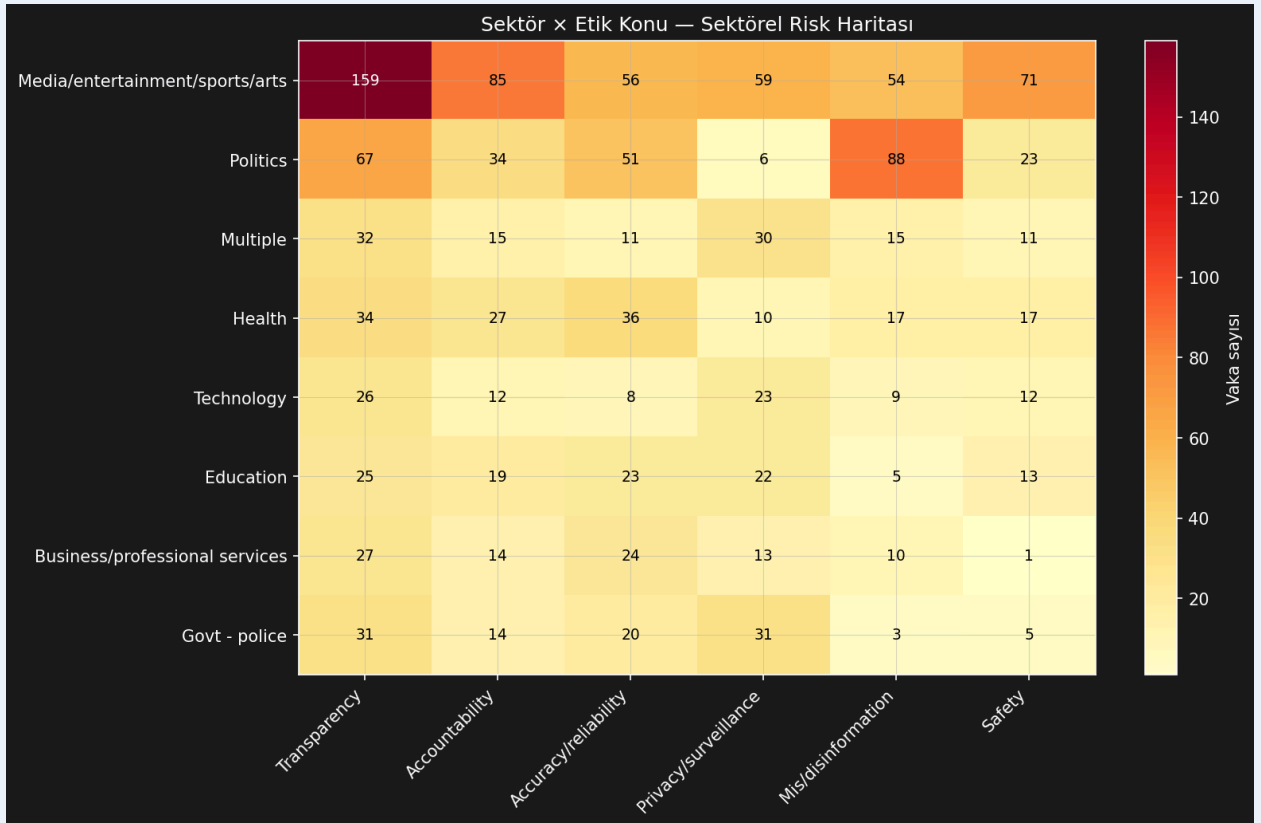


The main concern was also seen to shift over time. The share of privacy decreased, while the share of accountability increased sharply after 2024.



07_konu_evrimi.png

Different sectors were shown to carry different dominant risks. Disinformation stands out in politics, privacy in policing, and accuracy in health.



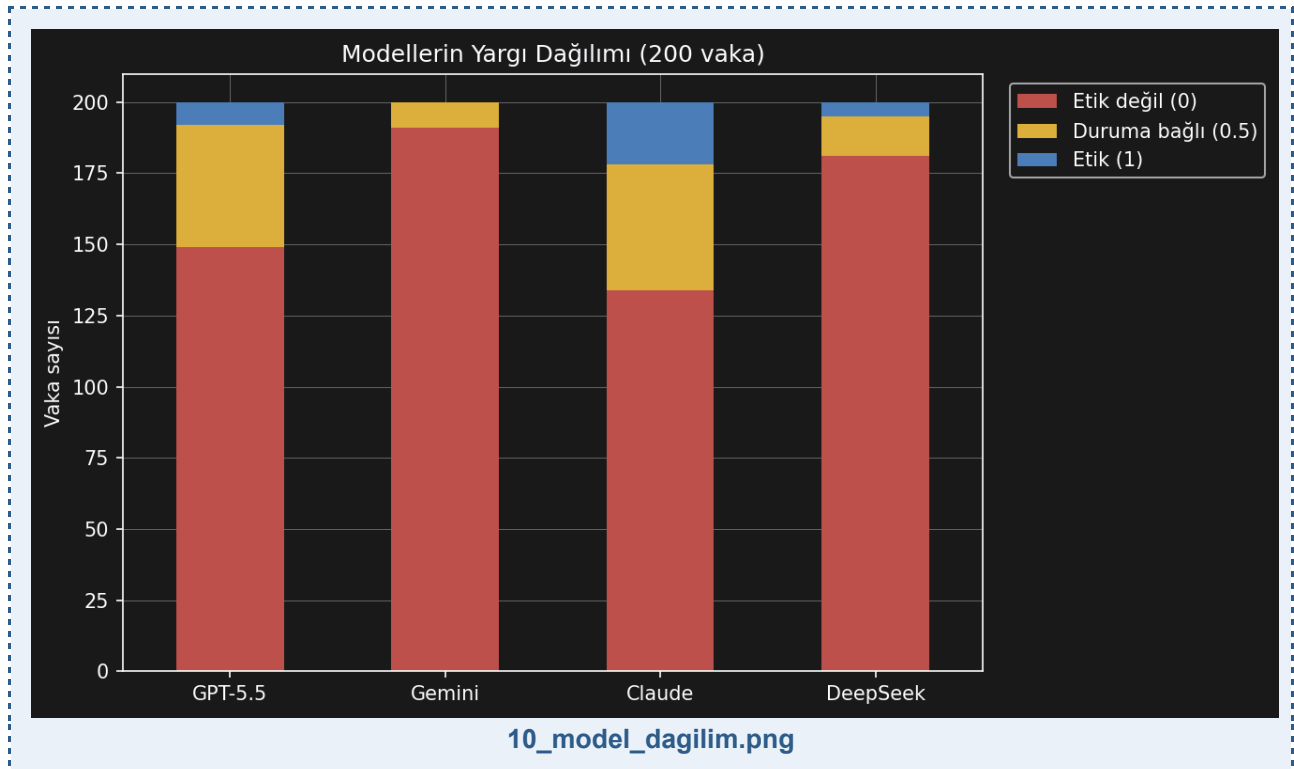
08_sektorel_risk.png

4. Phase 2. Comparison of AI Models

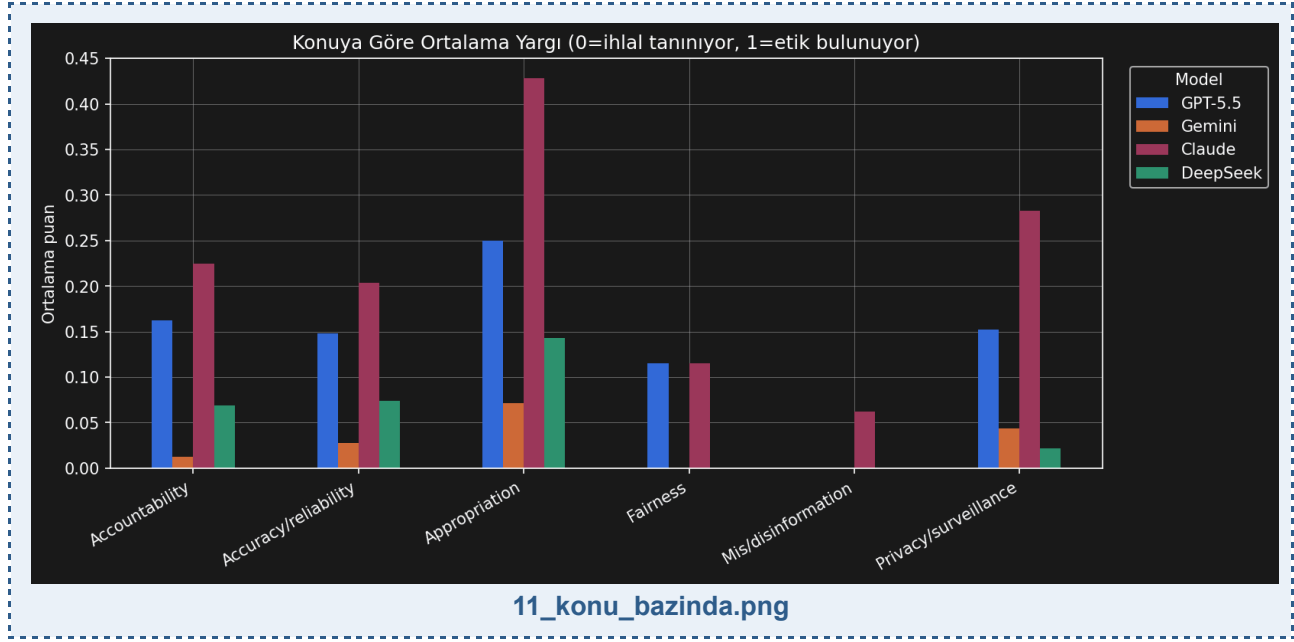
In the second phase, four large language models, GPT-5.5, Gemini, Claude, and DeepSeek, were compared. A sample of 200 real cases was used. For each case, every model was asked the same question, namely whether the described action is ethically acceptable. The answers were coded as 0 for not ethical, 0.5 for context dependent, and 1 for ethical. Because every case is a documented violation, an answer close to 0 is expected.

The main findings are summarised below.

All four models judged almost every case as not ethical, which indicates that violations are recognised reliably. The meaningful difference was found not in which model is more ethical, but in how each model handles uncertainty. Gemini was the strictest and rarely moved away from a clear judgement. DeepSeek was also strict and tended to avoid the middle option. GPT-5.5 and Claude were more cautious and used the context dependent answer more often.



The highest agreement between models was seen in disinformation and fairness. The widest disagreement appeared in appropriation, which covers copyright and data use, where Claude was the most flexible.



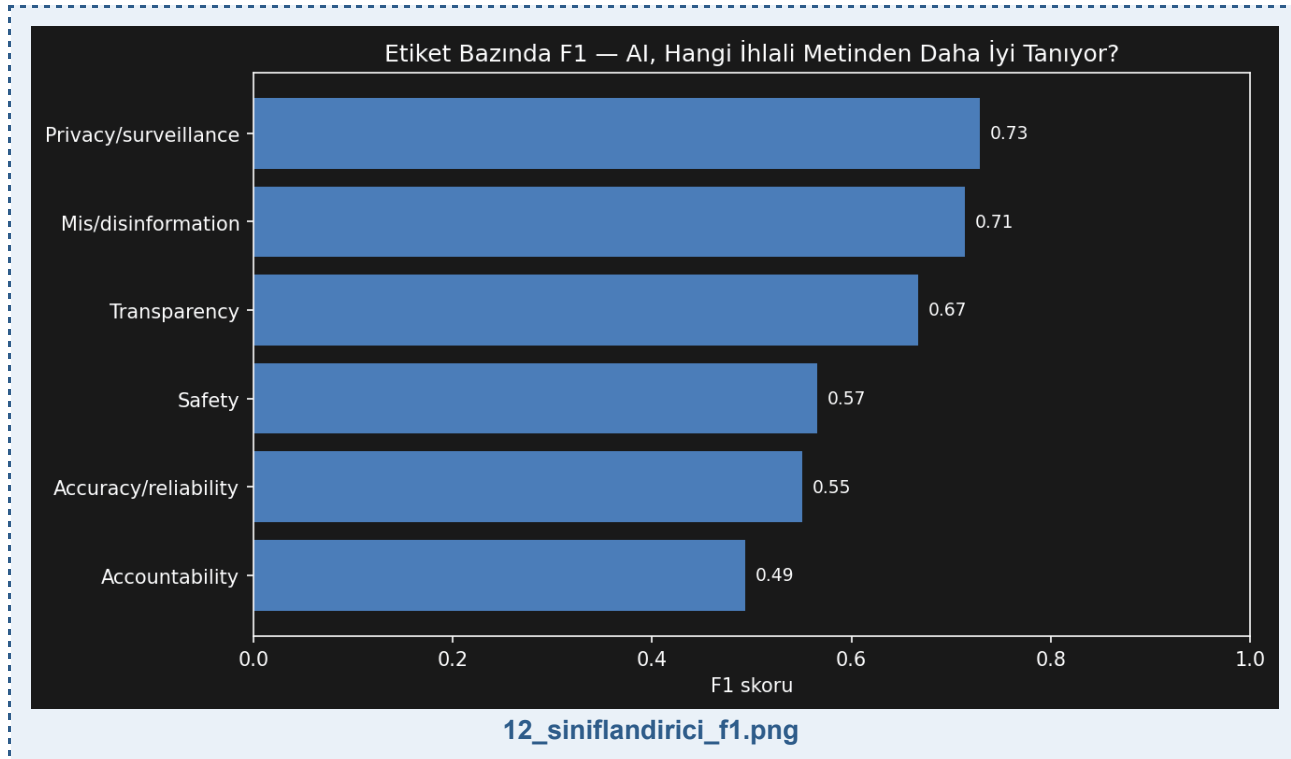
It should be noted that what the models state was measured, not what they actually do. A model may declare an action unethical and still perform it. This gap is treated as a limitation of the study.

5. Phase 3. A Machine Learning Model

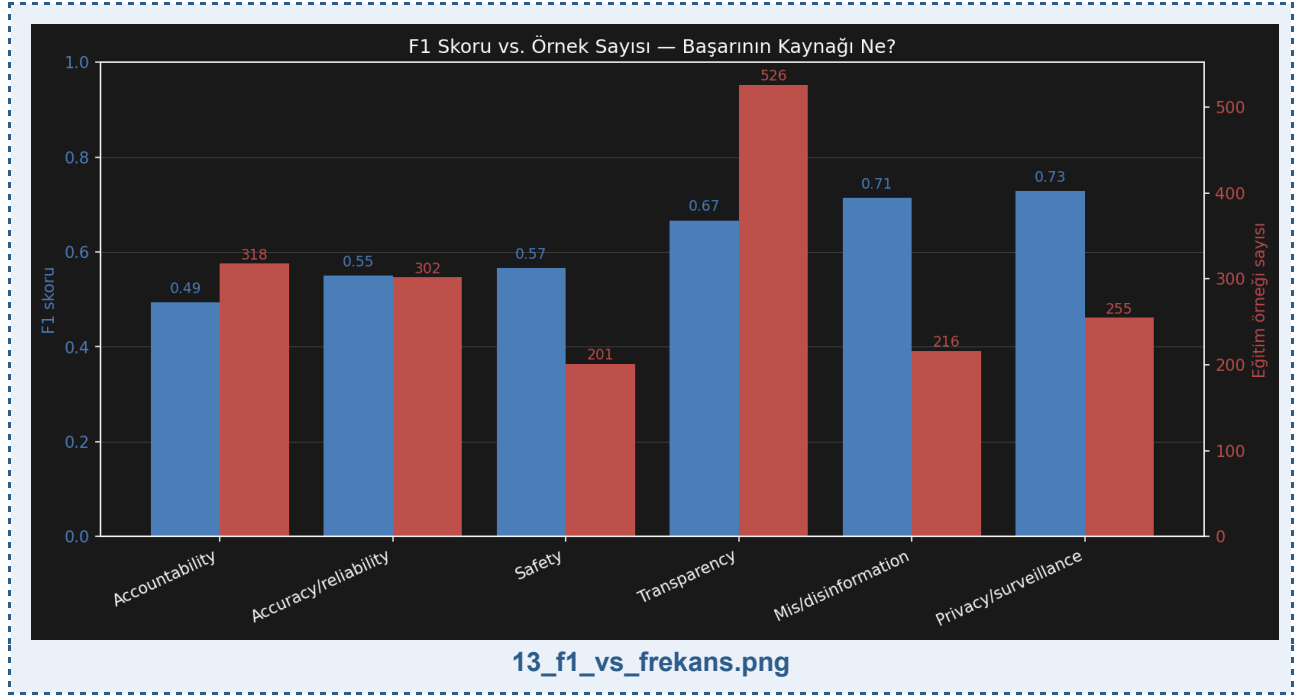
In the third phase, a small model was built to predict the ethical topic of a case directly from its title. A set of 915 cases and the six most frequent topics were used. Because cases often belong to more than one topic, a multi label approach was chosen. The model combines TF-IDF features with logistic regression.

The main findings are summarised below.

An overall Macro-F1 of 0.62 was reached, which is a strong result given that only short titles were available. Concrete topics were recognised well, with privacy at 0.73 and disinformation at 0.71, while the abstract topic of accountability remained low at 0.49.



The success was found to be independent of the amount of training data, since the correlation between the score and the number of examples was only 0.01. Performance is therefore driven by how concrete a topic is, not by how many examples exist.



Clear word patterns were also learned for each topic. Privacy was linked to terms such as facial recognition and surveillance, while disinformation was linked to terms such as deepfake and election.

6. Conclusion

The three phases point to a single conclusion. AI systems perform well on concrete and technical problems, such as surveillance and disinformation, where the harm is visible in the text. They are weaker, or divided, on abstract and normative problems, such as responsibility and fair use. For this reason, AI should not be treated as an ethical authority. It is better understood as an early warning tool for clear and concrete violations.

7. Limitations

- Only the stated judgement of each model was measured, not its actual behaviour.
- Each model was tested once, with a single version, in a single language.
- The data is a record of problems, so widely used developers appear more often without being more guilty.
- The topic labels were assigned by AIAIC editors, and a different taxonomy might give different results.
- The classifier used only the title of each case rather than the full description.